

```
import pandas as pd

df = pd.read_csv('/content/sales.csv', encoding='latin1')

df.head()
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	...	Postal Code	Region	Product ID
0	1	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	42420	South	FUR-BO-10001798
1	2	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	42420	South	FUR-CH-10000454
2	3	CA-2016-138688	06-12-2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	...	90036	West	OFF-LA-10000240
3	4	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	33311	South	FUR-TA-10000577
4	5	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	33311	South	OFF-ST-10000760

5 rows × 21 columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Row ID                9994 non-null  int64
1   Order ID              9994 non-null  object
2   Order Date            9994 non-null  object
3   Ship Date             9994 non-null  object
4   Ship Mode             9994 non-null  object
5   Customer ID           9994 non-null  object
6   Customer Name         9994 non-null  object
7   Segment              9994 non-null  object
8   Country               9994 non-null  object
9   City                 9994 non-null  object
10  State                9994 non-null  object
11  Postal Code           9994 non-null  int64
12  Region               9994 non-null  object
13  Product ID           9994 non-null  object
14  Category             9994 non-null  object
15  Sub-Category         9994 non-null  object
16  Product Name         9994 non-null  object
17  Sales                9994 non-null  float64
18  Quantity             9994 non-null  int64
19  Discount              9994 non-null  float64
20  Profit               9994 non-null  float64
dtypes: float64(3), int64(3), object(15)
memory usage: 1.6+ MB
```

```
df["Order Date"].dtype
```

```
dtype('O')
```

```
df["Sales"].sum()
```

```
np.float64(2297200.8603000003)
```

```
df.groupby("Region")["Sales"].sum()
```

```
df.groupby("Region")["Sales"].sum().sort_values(ascending=False).head()
```

Region	Sales
Central	501239.8908
East	678781.2400
South	391721.9050
West	725457.8245

dtype: float64

```
df.groupby("Customer Name")["Sales"].sum().sort_values(ascending=False).head()
```

Customer Name	Sales
Sean Miller	25043.050
Tamara Chand	19052.218
Raymond Buch	15117.339
Tom Ashbrook	14595.620
Adrian Barton	14473.571

dtype: float64

```
df["Order Date"] = pd.to_datetime(df["Order Date"], format='mixed')
```

```
df["Month"] = df["Order Date"].dt.to_period("M")
```

```
df.groupby("Month")["Sales"].sum()
```

	Sales
Month	
2014-01	14236.8950
2014-02	4519.8920
2014-03	55691.0090
2014-04	28295.3450
2014-05	23648.2870
2014-06	34595.1276
2014-07	33946.3930
2014-08	27909.4685
2014-09	81777.3508
2014-10	31453.3930
2014-11	78628.7167
2014-12	69545.6205
2015-01	18174.0756
2015-02	11951.4110
2015-03	38726.2520
2015-04	34195.2085
2015-05	30131.6865
2015-06	24797.2920
2015-07	28765.3250
2015-08	36898.3322
2015-09	64595.9180
2015-10	31404.9235

2015-11 75972.5635

```
df.to_csv("cleaned_sales.csv", index=False)
```

2016-01	18542.4910
2016-02	22978.8150
2016-03	51715.8750
2016-04	38750.0390
2016-05	56987.7280
2016-06	40344.5340
2016-07	39261.9630
2016-08	31115.3743
2016-09	73410.0249
2016-10	59687.7450
2016-11	79411.9658